

Hurdle and zero-inflated models

1 Exercise 1

A random variable Y is said to follow a negative binomial distribution, denoted as $\mathcal{NB}(\mu, \vartheta)$, when its probability mass function (p.m.f.) is

$$\mathbb{P}(Y = y; \vartheta, \mu) = \frac{\Gamma(y + \vartheta)}{\Gamma(\vartheta)\Gamma(y + 1)} \left(\frac{\vartheta}{\mu + \vartheta} \right)^\vartheta \left(\frac{\mu}{\mu + \vartheta} \right)^y,$$

where $\Gamma(u) = \int_0^\infty t^{u-1} \exp(-t) dt$ denotes the gamma function, $\mu > 0$, $\vartheta > 0$ and $y = 0, 1, 2, \dots$

- a) Give the p.m.f. of a random variable distributed as a truncated $\mathcal{NB}(\mu, \vartheta)$, *i.e.* with strictly positive outcomes.
- b) Define a hurdle model for responses issued from a $\mathcal{NB}$ distribution with excess of zeros. Write the log-likelihood function of that model.
- c) Show that, in a similar fashion as for the Poisson hurdle model, the log-likelihood function computed in **b)** can be separated into two components: the first one for the peak at 0 and the second one for the strictly positive outcomes.
 - (a) What does it imply for the estimation process?
 - (b) What does it imply from the point of view of statistical inference?

2 Exercise 2

Assume that $Y_1, \dots, Y_n$ are responses for n individuals issued from a zero-inflated Poisson model (ZIP) with covariates $x_1, \dots, x_n$ and parameter α explaining zero responses; and covariates $z_1, \dots, z_n$ and parameter δ explaining positive responses.

Let W_i be an unobservable binary variable that indicates the risk state of individual i : $W_i = 1$ if the subject is not at risk of an event; and $W_i = 0$ if the subject is at risk. Notice that Y_i denote the event count for a subject and this event occurs only when $W_i = 0$. We also suppose that $\Pr\{W_i = 1\} = \pi_i$ (where π_i is a function of $x_i^T \alpha$) and

$P\{Y_i = y|W_i = 0\} = \lambda_i^y e^{-\lambda_i} / y!$ (where λ_i is a function of $z_i^T \delta$) i.e. $Y_i|W_i = 0$ is issued from the Poisson distribution.

a) Write the likelihood function

$$L(\alpha, \delta | y_1, \dots, y_n, w_1, \dots, w_n) = \prod_{i=1}^n P\{Y_i = y_i | W_i = w_i\}.$$

Show that the corresponding log-likelihood function $\ell(\alpha, \delta | y_1, \dots, y_n, w_1, \dots, w_n)$ can be written as

$$\ell(\alpha | y_1, \dots, y_n, w_1, \dots, w_n) + \ell(\delta | y_1, \dots, y_n, w_1, \dots, w_n)$$

i.e. the parameters explaining the presence of zeros can be separated from the other parameters. What does it imply numerically?

In practice the values $w_1, \dots, w_n$ are never observed and the estimators $\hat{\alpha}, \hat{\delta}$ are obtained with the EM-algorithm¹.

With the EM algorithm $W_1, \dots, W_n$ are considered to be missing and are estimated by their expected values $w_1^{(k)}, \dots, w_n^{(k)}$ under the current estimates $\alpha^{(k)}$ and $\delta^{(k)}$ (E-step). Then $\ell(\alpha, \delta | y_1, \dots, y_n, w_1^{(k)}, \dots, w_n^{(k)})$ is maximized (M-step), and the E- and M-steps are iterated until convergence.

More precisely, assuming $\alpha^{(k)}, \delta^{(k)}, w_1^{(k)}, \dots, w_n^{(k)}$ to be the values obtained at the (k) -th iteration, the iteration $(k+1)$ of the EM-algorithm requires the following two steps.

- *E-step.* Find $w_i^{(k+1)}$ the conditional expectation of W_i given $y_i, \alpha^{(k)}, \delta^{(k)}$ i.e. the expectation of W_i with respect to the following probability mass function $P\{W_i | Y_i = y_i, \alpha^{(k)}, \delta^{(k)}\}$.
- *M-step.* Find $\alpha^{(k+1)}, \delta^{(k+1)}$ as $\arg\max_{\alpha, \delta} \ell(\alpha, \delta | y_1, \dots, y_n, w_1^{(k)}, \dots, w_n^{(k)})$.

We would like you to provide the detailed description of the E-step, by answering the following questions.

¹Dempster, A.P., Laird, N.M., and Rubin, D.B. (1977) Maximum Likelihood Estimation from Incomplete Data via the EM Algorithm, JRSS-B

- b) Find the probability $P\{W_i|Y_i = y_i, \alpha^{(k)}, \delta^{(k)}\}$, for short you can write it as $P\{W_i|y_i, \alpha, \delta\}$. In other words find $P\{W_i = 1|Y_i = 0, \alpha^{(k)}, \delta^{(k)}\}$, $P\{W_i = 0|Y_i = 0, \alpha^{(k)}, \delta^{(k)}\}$ and $P\{W_i = 0|Y_i = y_i > 0, \alpha^{(k)}, \delta^{(k)}\}$.

Hint: use the Bayes formula $P\{H|D\} = \frac{P\{D|H\}P\{H\}}{P\{D\}}$.

- c) Using $P\{W_i|Y_i, \alpha^{(k)}, \delta^{(k)}\}$ find $w_i^{(k+1)} = E(W_i|Y_i = y_i, \alpha^{(k)}, \delta^{(k)})$.
- d) How can we interpret the values of $w_1^{(k+1)}, \dots, w_n^{(k+1)}$ obtained at the last iteration of the algorithm ?

3 Exercise 3

Health surveys are often used to study health care expenditures and especially the impact of income and insurance level on the frequency of health care services use. Here, we are interested to see if a high insurance level (with high premium) “causes” individuals to consume more health care services.

Import the `docvisits.asc` data file, which is part of the 1977-1978 Australian Health Survey and contains information on the number of consultations with a doctor or specialist, denoted `dvisits`, in the two-weeks period before an interview. These counts represent our response variable and we consider the following thirteen explanatory variables:

- `sex` coded 1 for female,
- `age` in years/100,
- `agesq`, $(\text{age})^2/1000$,
- `income`, annual income/1000,
- `illness`, number of illnesses in past two weeks, with five or more coded as 5,
- `actdays`, number of reduced activity days in past two weeks due to illness or injury,
- `hscore`, general health questionnaire score, with high scores indicating bad health,
- `chcond1` coded 1 if chronic condition(s) but not limited in activity,

- `chcond2` coded 1 if chronic condition(s) and limited in activity,
 - `levyplus`, `freepoor` and `freerepa`, three dummy variables for levels of health insurance, where “levyplus” represents a higher level of insurance cover while “freepoor” and “freerepa” are basic levels of state-provided insurances.
- a) Source the `truncpoisson.R` code available from the course website to create the `truncpoisson` GLM family object. Fit manually the two steps of the Poisson hurdle model to the data by using the `family=truncpoisson` argument in the `glm` command. Compare your results to the output from the `hurdle` command of the `pscl` package.
- b) Fit a simple Poisson GLM and a ZIP model. Among the three models, which provides the best fit?
- c) Do high insurance levels make the number of consultations increase?